Esempio 1. Risolvere equazione alle differenze ($\mathbb{T} = \mathbb{Z}$)

$$\Delta^2 (\Delta + 1)^2 (\Delta^2 + 1)\, y(t) = \delta_0(t-3) + (-1)^t t^2 + t$$

$$\begin{aligned}
y_{g,o}(t) &= c_1\,\delta_0(t) + c_2\,\delta_0(t-1) + c_3\,(-1)^t + c_4\,(-1)^t\,t + c_5 \cos\left(\frac{\pi}{2}\,t\right) + c_6 \sin\left(\frac{\pi}{2}\,t\right)\\
y_{p,\mathrm{nr}}(t) &= \alpha_1\,\delta_0(t) + \alpha_2\,\delta_0(t-1) + \alpha_3\,\delta_0(t-2) + \alpha_4\,\delta_0(t-3) + \alpha_5\,(-1)^t + \alpha_6\,(-1)^t\,t + \alpha_7(-1)^t\,t^2 + \alpha_8 + \alpha_9\,t\\
y_p(t) &= \alpha_1\,\cancel{\delta_0}(t) + \alpha_2\,\delta_0(\cancel{t}-1) + \alpha_3\,\delta_0(t-2) + \alpha_4\,\delta_0(t-3) + \alpha_5\,\delta_0(t-4) + \alpha_6\,\delta_0(t-5) + \alpha_7\,(\cancel{-}1)^t + \alpha_8\,(\cancel{-}1)^t\,t + \alpha_9(-1)^t\,t^2 + \alpha_{10}\,(-1)^t\,t^3 +\\
&\quad \alpha_{11}\,(-1)^t\,t^4 + \alpha_{12} + \alpha_{13}\,t
\end{aligned}$$

$$\begin{aligned}
\Delta^2 (\Delta + 1)^2 (\Delta^2 + 1)\, y(t) &= \delta_0(t) + (-1)^t + t\\
y_{g,o}(t) &= c_1\,\delta_0(t) + c_2\,\delta_0(t-1) + c_3\,(-1)^t + c_4\,(-1)^t\,t + c_5 \cos\left(\frac{\pi}{2}\,t\right) + c_6 \sin\left(\frac{\pi}{2}\,t\right)\\
y_{p,\mathrm{nr}}(t) &= \alpha_1\,\delta_0(t) + \alpha_2\,(-1)^t + \alpha_3 + \alpha_4\,t\\
y_p(t) &= \alpha_1\,\delta_0(t) + \alpha_2\,\delta_0(t-1) + \alpha_3\,\delta_0(t-2) + \alpha_4\,(-1)^t + \alpha_5\,(-1)^t\,t + \alpha_6\,(-1)^t\,t^2 + \alpha_7 + \alpha_8\,t
\end{aligned}$$

$$\begin{aligned}
\dot{x} &= A\,x + B\,u\\
A &= \begin{pmatrix} -1 & 1 & 2\\ 0 & -1 & 1\\ 0 & 0 & -1 \end{pmatrix}\\
B &= \begin{pmatrix} 0\\ 0\\ 1 \end{pmatrix}\\
x_0 &= \begin{pmatrix} 1\\ 0\\ 0 \end{pmatrix}\\
u(t) &= \mathrm{e}^{-t}\,\delta_{-1}(t)
\end{aligned}$$

$$\begin{aligned}
\det(s\,I - A) &= (s+1)^3\\
s &= -1\,\mathrm{con\,molt.}3\\
&\ \mathrm{modi:}\,\mathrm{e}^{-t}, t\,\mathrm{e}^{-t}, t^2\,\mathrm{e}^{-t}
\end{aligned}$$

$$\begin{aligned}
x(s) &= (s\,I - A)^{-1}\,x_0 + (s\,I - A)^{-1}\,B\,u(s)\\
(s\,I - A)^{-1} &= \begin{pmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} & \frac{2}{(s+1)^2} + \frac{1}{(s+1)^3}\\ 0 & \frac{1}{s+1} & \frac{1}{(s+1)^2}\\ 0 & 0 & \frac{1}{s+1} \end{pmatrix}\\
\mathrm{e}^{A\,t} &= \begin{pmatrix} \mathrm{e}^{-t} & t\,\mathrm{e}^{-t} & 2\,t\,\mathrm{e}^{-t} + \frac{t^2}{2}\,\mathrm{e}^{-t}\\ 0 & \mathrm{e}^{-t} & t\,\mathrm{e}^{-t}\\ 0 & 0 & \mathrm{e}^{-t} \end{pmatrix}\\
(s\,I - A)^{-1}\,B &= \begin{pmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} & \frac{2\,(s+1)+1}{(s+1)^3}\\ 0 & \frac{1}{s+1} & \frac{1}{(s+1)^2}\\ 0 & 0 & \frac{1}{s+1} \end{pmatrix} \cdot \begin{pmatrix} 0\\ 0\\ 1 \end{pmatrix}\\
&= \begin{pmatrix} \frac{2\,(s+1)+1}{(s+1)^3}\\ \frac{1}{(s+1)^2}\\ \frac{1}{s+1} \end{pmatrix}\\
u(s) &= \frac{1}{s+1}\\
x(s) &= \begin{pmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} & \frac{2\,(s+1)+1}{(s+1)^3}\\ 0 & \frac{1}{s+1} & \frac{1}{(s+1)^2}\\ 0 & 0 & \frac{1}{s+1} \end{pmatrix} \cdot \begin{pmatrix} 1\\ 0\\ 0 \end{pmatrix} + \begin{pmatrix} \frac{2\,(s+1)+1}{(s+1)^3}\\ \frac{1}{(s+1)^2}\\ \frac{1}{s+1} \end{pmatrix} \frac{1}{s+1}\\
&= \begin{pmatrix} \frac{1}{s+1} + \frac{2\,(s+1)+1}{(s+1)^4}\\ \frac{1}{(s+1)^3}\\ \frac{1}{(s+1)^2} \end{pmatrix}
\end{aligned}$$

$$
\begin{aligned}
x_l &= \begin{pmatrix} \frac{1}{s+1} \\ 0 \\ 0 \end{pmatrix}\\
x_l(t) &= \begin{pmatrix} \mathrm{e}^{-t} \\ 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
x_f(t) &= \begin{pmatrix} \frac{2\,(s+1)+1}{(s+1)^4} \\ \frac{1}{(s+1)^3} \\ \frac{1}{(s+1)^2} \end{pmatrix}\\
&= \begin{pmatrix} \frac{2}{(s+1)^3}+\frac{1}{(s+1)^4} \\ \frac{1}{(s+1)^3} \\ \frac{1}{(s+1)^2} \end{pmatrix}\\
&= \begin{pmatrix} 2\,\frac{t^2}{2!}\,\mathrm{e}^{-t}+\frac{t^3}{3!}\,\mathrm{e}^{-t} \\ \frac{t^2}{2!}\,\mathrm{e}^{-t} \\ t\,\mathrm{e}^{-t} \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
&= \begin{pmatrix} \mathrm{partfrac}\left(\frac{1}{s+1}+\frac{2\,(s+1)+1}{(s+1)^4}, s\right) \\ \frac{1}{(s+1)^3} \\ \frac{1}{(s+1)^2} \end{pmatrix}\\
&= \begin{pmatrix} \frac{1}{s+1}+\frac{2}{(s+1)^3}+\frac{1}{(s+1)^4} \\ \frac{1}{(s+1)^3} \\ \frac{1}{(s+1)^2} \end{pmatrix}\\
x(t) &= \begin{pmatrix} \mathrm{e}^{-t}+2\,\frac{t^2}{2!}\,\mathrm{e}^{-t}+\frac{t^3}{3!}\,\mathrm{e}^{-t} \\ \frac{t^2}{2!}\,\mathrm{e}^{-t} \\ t\,\mathrm{e}^{-t} \end{pmatrix}\\
&= \underbrace{\begin{pmatrix} \mathrm{e}^{-t}+2\,\frac{t^2}{2!}\,\mathrm{e}^{-t} \\ \frac{t^2}{2!}\,\mathrm{e}^{-t} \\ t\,\mathrm{e}^{-t} \end{pmatrix}}_{\text{pseudo}-\text{transitoria}} + \underbrace{\begin{pmatrix} \frac{t^3}{3!}\,\mathrm{e}^{-t} \\ 0 \\ 0 \end{pmatrix}}_{\text{pseudo}-\text{permanente}}
\end{aligned}
$$

$$
\begin{aligned}
x(t+1) &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} x + \begin{pmatrix} 1 \\ -1 \end{pmatrix} t\\
\det(z\,I-A) &= (z-1)\,z\\
&\quad \text{modi}: 1, \delta_0(t)
\end{aligned}
$$

$$
x_p(t) = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} t + \begin{pmatrix} \alpha_5 \\ \alpha_6 \end{pmatrix} t^2
$$

$$
\begin{aligned}
A^t &= \begin{pmatrix} \delta_0(t) & 0 \\ 0 & 1^t \end{pmatrix}\\
\mathrm{subst}\left(t=t+1, \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} t + \begin{pmatrix} \alpha_5 \\ \alpha_6 \end{pmatrix} t^2\right) &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} . \left(\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} t + \begin{pmatrix} \alpha_5 \\ \alpha_6 \end{pmatrix} t^2\right) + \begin{pmatrix} 1 \\ -1 \end{pmatrix} t\\
\begin{pmatrix} \alpha_5\,(t+1)^2+\alpha_3\,(t+1)+\alpha_1 \\ \alpha_6\,(t+1)^2+\alpha_4\,(t+1)+\alpha_2 \end{pmatrix} &= \begin{pmatrix} t \\ \alpha_6\,t^2+\alpha_4\,t-t+\alpha_2 \end{pmatrix}\\
\alpha_6\,t^2+2\,\alpha_6\,t+\alpha_4\,t+\alpha_6+\alpha_4+\alpha_2 &= \alpha_6\,t^2+\alpha_4\,t-t+\alpha_2\\
\alpha_5 &= 0\\
\alpha_3 &= 1\\
\alpha_1 &= -1\\
\alpha_6 &= \alpha_6 = -\frac{1}{2}\\
2\,\alpha_6+\alpha_4 &= \alpha_4-1\\
\alpha_6+\alpha_4+\cancel{\alpha_2} &= \cancel{\alpha_2}\\
\alpha_4 &= \frac{1}{2}
\end{aligned}
$$

$$\alpha_2 \quad \text{è} \quad \text{arbitrario}$$

$$\begin{aligned}
A^t\, c &= \begin{pmatrix} \delta_0(t) & 0 \\ 0 & 1^t \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\
x_{g,o}(t) &= \begin{pmatrix} c_1\, \delta_0(t) \\ c_2 \end{pmatrix} \\
x_p(t) &= \begin{pmatrix} -1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} t + \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} t^2
\end{aligned}$$

$$\begin{aligned}
\Delta\,(\Delta-2)\,(\Delta-1)\,y &= 2^t \\
y_{g,o}(t) &= c_1\,\delta_0(t) + c_2\,2^t + c_3 \\
y_{p,\text{nr}} &= \alpha_1\,2^t \\
y_p(t) &= \cancel{\alpha_1\,2^t} + \alpha_2\,2^t\,t
\end{aligned}$$

$$\begin{aligned}
p(\Delta) &= \Delta^3 - 3\,\Delta^2 + 2\,\Delta \\
y_p(t+3) - 3\,y_p(t+2) + 2\,y_p(t+1) &= 2^t \\
\alpha_1\,\cancel{2^t}\,2^3\,(t+3) - 3\,\alpha_1\,\cancel{2^t}\,4\,(t+2) + 2\,\alpha_1\,\cancel{2^t}\,2\,(t+1) &= \cancel{2^t} \\
\alpha_1\,2^3\,(t+3) - 3\,\alpha_1\,4\,(t+2) + 2\,\alpha_1\,2\,(t+1) &= 1 \\
4\,\alpha_1 &= 1 \\
\alpha_1 &= \frac{1}{4}
\end{aligned}$$

$$y_p(t) = \frac{1}{4}\,2^t\,t$$

$$\begin{aligned}
\dot{x} &= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} x + \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ t \end{pmatrix} \\
&= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} x + \begin{pmatrix} t+1 \\ -t \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
p(s) &= (s-1)^2 - 4 \\
&= s^2 - 2\,s - 3 \\
s_{1,2} &= 1 \pm \sqrt{1+3} \\
&= \{3, -1\} \\
&\ \text{modi}\colon \mathrm{e}^{3t}, \mathrm{e}^{-t}
\end{aligned}$$

$$\begin{aligned}
\frac{1}{p(s)} &= \text{partfrac}\!\left(\frac{1}{s^2 - 2\,s - 3}, s\right) \\
&= \frac{1}{4\,(s-3)} - \frac{1}{4\,(s+1)} \\
f_1(s) &= \frac{1}{4\,(s\cancel{-3})} \frac{(s\cancel{-3})\,(s+1)}{p(s)} \\
f_2(s) &= -\frac{1}{4\,(s\cancel{+1})} \frac{(s-3)\,(s\cancel{+1})}{p(s)}
\end{aligned}$$

$$\begin{aligned}
e_1(s) &= \frac{1}{4}\,(s+1) \quad \lambda_1 = 3 \\
e_2(s) &= -\frac{1}{4}\,(s-3) \quad \lambda_2 = -1
\end{aligned}$$

$$\begin{aligned}
E_1 &= \frac{1}{4}\left(\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} + \text{ident}(2)\right) \\
&= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \\
E_2 &= -\frac{1}{4}\left(\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} - 3\,\text{ident}(2)\right) \\
&= \begin{pmatrix} \frac{1}{2} & -(\frac{1}{2}) \\ -(\frac{1}{2}) & \frac{1}{2} \end{pmatrix} \\
A &= \lambda_1\,E_1 + \lambda_2\,E_2 \\
&= 3 \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} + (-1) \begin{pmatrix} \frac{1}{2} & -(\frac{1}{2}) \\ -(\frac{1}{2}) & \frac{1}{2} \end{pmatrix}
\end{aligned}$$

$$
\begin{aligned}
&= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \\
\mathrm{e}^{At} &= \mathrm{e}^{3t}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} + \mathrm{e}^{-t}\begin{pmatrix} \frac{1}{2} & -\left(\frac{1}{2}\right) \\ -\left(\frac{1}{2}\right) & \frac{1}{2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} \\ \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} \end{pmatrix} \\
x_{g,o}(t) &= \mathrm{e}^{At} c \\
x_p(t) &= \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} t \\
\begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} &= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \cdot \left(\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} t \right) + \begin{pmatrix} t+1 \\ -t \end{pmatrix} \\
\begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} &= \begin{pmatrix} 2\,\alpha_4\, t + \alpha_3\, t + t + 2\,\alpha_2 + \alpha_1 + 1 \\ \alpha_4\, t + 2\,\alpha_3\, t - t + \alpha_2 + 2\,\alpha_1 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
2\,\alpha_4 + \alpha_3 + 1 &= 0 \\
2\,\alpha_2 + \alpha_1 + 1 &= \alpha_3 \\
\alpha_4 + 2\,\alpha_3 - 1 &= 0 \\
\alpha_4 &= \alpha_2 + 2\,\alpha_1
\end{aligned}
$$

$$
\begin{aligned}
& \mathrm{solve}([2\,\alpha_4 + \alpha_3 + 1, 2\,\alpha_2 + \alpha_1 + 1 - \alpha_3, \alpha_4 + 2\,\alpha_3 - 1, \alpha_2 + 2\,\alpha_1 - \alpha_4]) \\
& \left[\left[\alpha_2 = \frac{1}{3}, \alpha_1 = -\left(\frac{2}{3}\right), \alpha_4 = -1, \alpha_3 = 1\right]\right] \\
x_p(t) &= \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} t
\end{aligned}
$$

$$
\begin{aligned}
x_{g,\mathrm{no}}(t) &= \begin{pmatrix} \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} \\ \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} t \\
x_0 &= \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} \\
\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} &= x_0 - \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
x(t) &= \underbrace{\begin{pmatrix} \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} \\ \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} \end{pmatrix} \left(x_0 - \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} \right)}_{\text{pseudo}-\text{transitoria}} + \underbrace{\begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} t}_{\text{pseudo}-\text{permanente}} \\
&= \underbrace{\begin{pmatrix} \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} \\ \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} \end{pmatrix} x_0}_{\text{libera}} - \underbrace{\begin{pmatrix} \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} \\ \frac{\mathrm{e}^{3t}}{2} - \frac{\mathrm{e}^{-t}}{2} & \frac{\mathrm{e}^{3t}}{2} + \frac{\mathrm{e}^{-t}}{2} \end{pmatrix} \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix}}_{\text{forzata}}
\end{aligned}
$$